

4.1 The structure of the Earth and the distribution of earthquakes and volcanoes

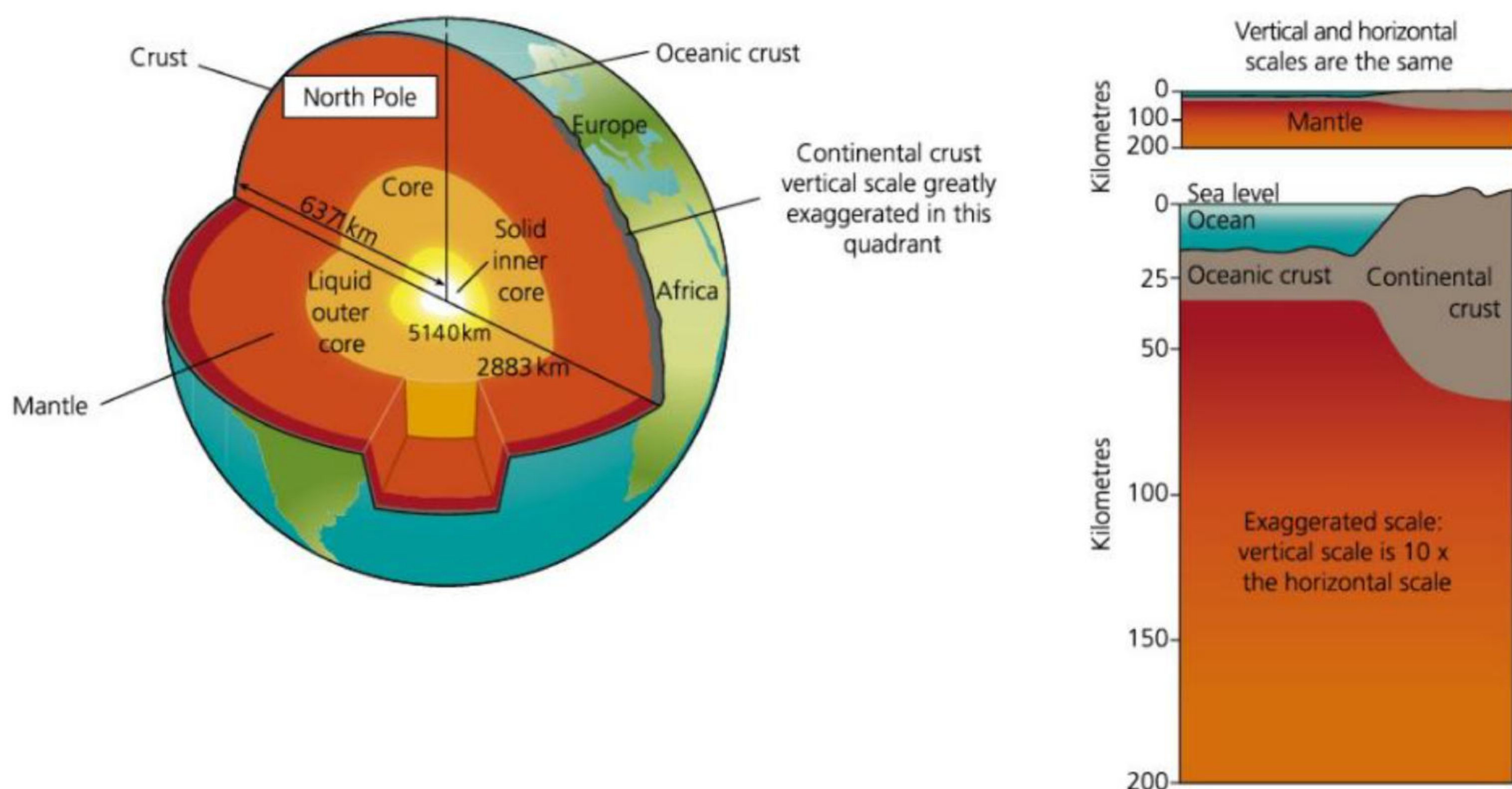
This chapter will explain:

- ★ the characteristics of the layers of the Earth, namely the inner core, outer core, mantle, crust and lithosphere
- ★ the main tectonic plates and how they move
- ★ the main types of plate boundary and how they relate to earthquakes and volcanoes.

The characteristics of the layers of the Earth

There are four main layers within the Earth ([Figure 4.1](#)):

- » The core is divided into two parts:
 - The inner core is solid. It is five times denser than surface rocks.
 - The outer core is semi-molten.
- » The mantle is semi-molten and about 2900 km thick.
- » The crust is solid and its depth varies between 10 km and 70 km. It is divided into two main types:
 - Continental crust: Mostly formed of granite and less dense than the oceanic crust.
 - Oceanic crust: Mostly formed of basalt. As it is denser than the continental crust, it plunges beneath it when they come together.
- » The lithosphere is the solid outer part of the Earth and includes the brittle upper part of the mantle and the crust.



▲ **Figure 4.1** The structure of the Earth

Activities

- 1 Compare the oceanic crust with the continental crust.
- 2 Compare the inner core with the outer core.

Tectonic plates and their movement

[Figure 4.2](#) shows the location of tectonic plates and their direction of movement relative to one another. When the plates collide, they produce tectonic activity and new landforms. There are seven large plates (five of which carry continents) and a number of smaller plates:

- » The main plates are the Pacific, Indo-Australian, Antarctic, North American, South American, African and Eurasian Plates.
- » Smaller plates include the Caribbean, Iranian, Arabian and Juan de Fuca Plates.

Tectonic plates are constantly moving. Two main mechanisms for plate movement have been identified:

- » Convection currents
- » Gravitational sliding.

Convection currents

Convection currents are partly responsible for the movement of tectonic plates. Heat produced by radioactive decay from the Earth's core causes rock (**magma**) to rise towards the crust and spread ([Figure 4.3](#)). The magma cools and sinks. This process is repeated in a cycle and is responsible for the movement of crustal plates.

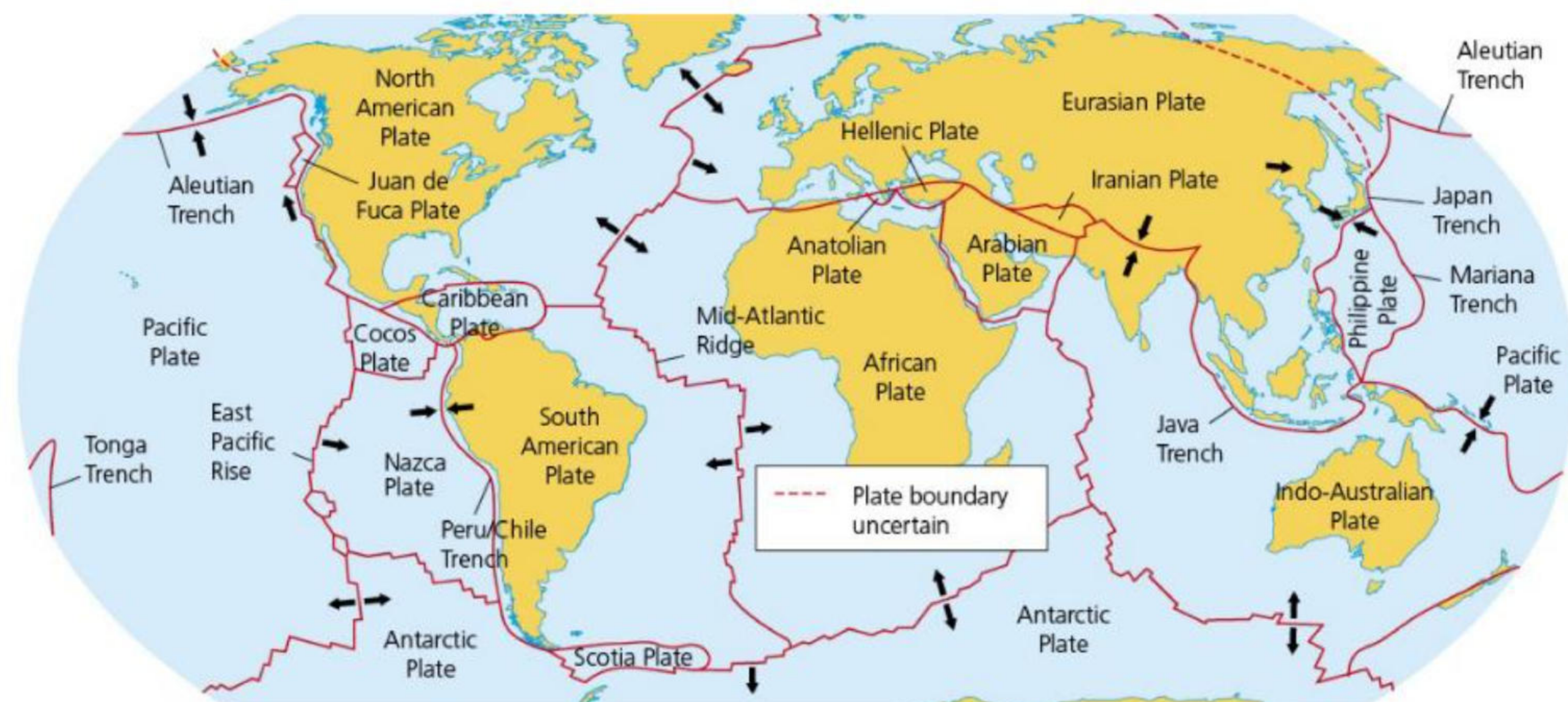
Gravitational sliding

It is now believed that gravity is also important in driving tectonic plate movement. There are two ways in which gravitational sliding operates ([Figure 4.3](#)):

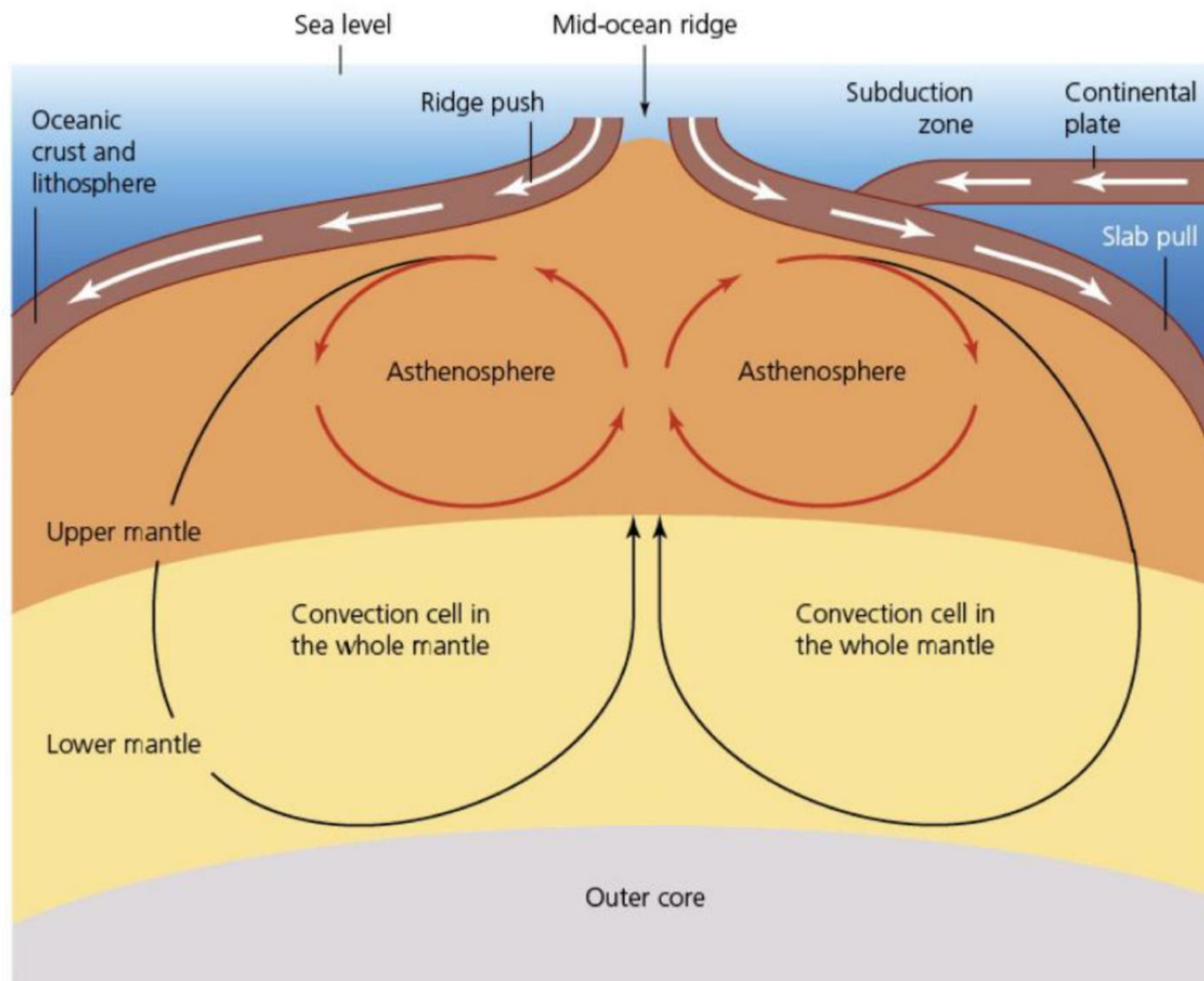
- » Ridge push: At mid-ocean ridges, new oceanic crust is pushed apart by rising magma from the mantle. As the mid-ocean ridge is higher than the ocean floor, gravity causes the plates to move downwards.
- » Slab pull: Gravity acting upon the colder and denser subducting (plunging) plate causes it to sink into the mantle and pull the remaining plate behind it.

Activities

- 1 Explain the difference between slab pull and ridge push.
 - 2 Identify one oceanic plate, one continental plate and one small plate.
-



▲ **Figure 4.2** The world's main tectonic plates



▲ **Figure 4.3** Convection currents, ridge push and slab pull

Types of plate boundaries

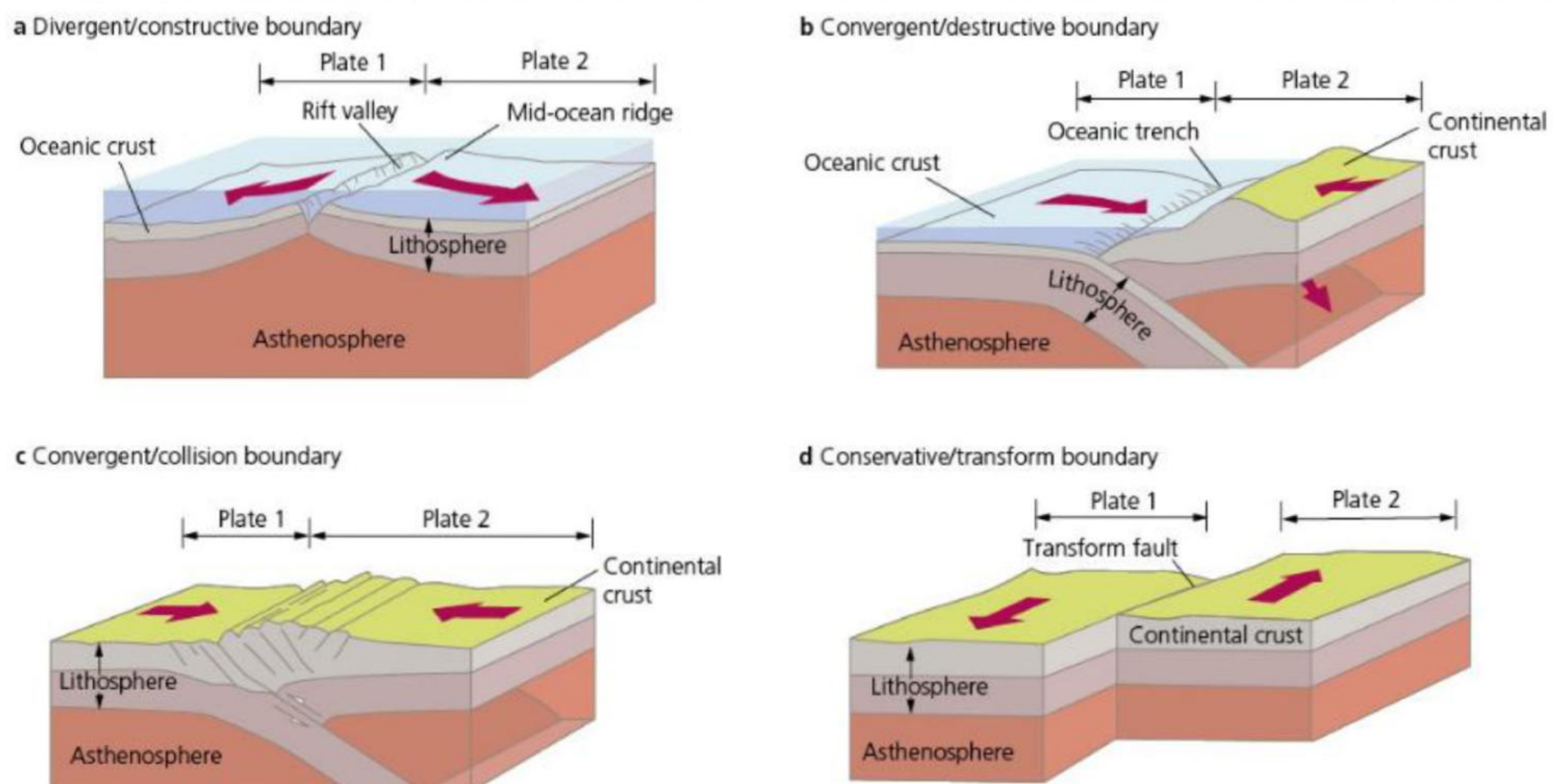
There are a number of different types of plate boundaries ([Table 4.1](#) and [Figure 4.4](#)). These include:

- » **divergent/constructive boundaries**, in which new oceanic crust is created
- » **convergent/destructive boundaries**, in which older crust is destroyed
- » **convergent/collision boundaries**, where plates are folded and crumpled
- » **conservative/transform boundaries**, where plates slip past each other, causing earthquakes to occur.

Different plate boundaries are associated with different tectonic activities: volcanic eruptions, folding and earthquake activity ([Figure 4.5](#)).

▼ **Table 4.1** The types of plate boundary

Divergent/constructive boundary	● Two plates move apart from each other, causing sea floor spreading. ● New oceanic crust is formed, creating mid-ocean ridges. ● Volcanic activity is common, e.g. the Mid-Atlantic Ridge (Europe is moving away from North America).
Convergent/destructive boundary	● The oceanic crust moves towards the continental crust and sinks beneath it due to its greater density. ● Deep-sea trenches and island arcs are formed. ● The continental crust is folded into fold mountains. ● Volcanic activity is common, e.g. the Nazca Plate is sinking under the South American Plate.
Convergent/collision boundary	● Two continental crusts collide. ● As neither can sink, they are folded up into fold mountains, e.g. the Indian Plate collided with the Eurasian Plate to form the Himalayas.
Conservative/transform boundary	● Two plates slip sideways past each other, but land is neither destroyed nor created, e.g. the San Andreas Fault in California.



▲ **Figure 4.4** The types of plate boundary



▲ **Figure 4.5** Tectonic activities **a** A folded landscape, Himalaya foothills **b** Thingvellir Rift Valley, Iceland **c** Volcanic eruption of Soufrière, Montserrat, with the former capital city Plymouth in the foreground **d** Tourists standing by the boiling mud springs, Soufriere, St Lucia

Activities

- 1 Name the type of plate boundary located:
 - a off the west coast of central America
 - b in the south Atlantic Ocean
 - c where the Indian Plate meets the Eurasian Plate.
- 2 Using [Figure 4.5 a-c](#), identify one location of:
 - a convergent/collision boundary
 - b a divergent/constructive plate boundary

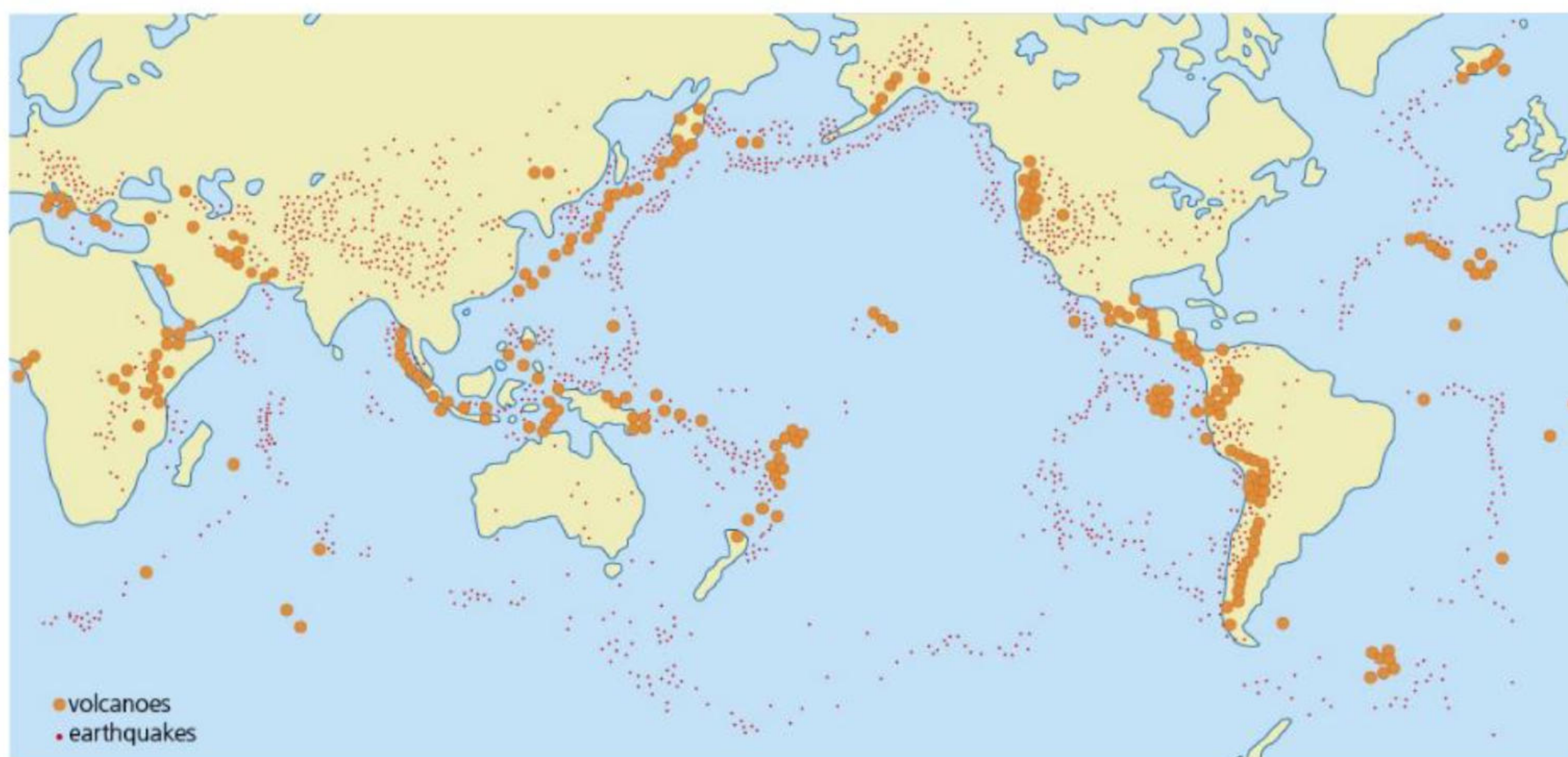
The location of earthquakes

Earthquakes are located very unevenly around the world ([Figure 4.6](#)). About 500,000 earthquakes are detected each year by sensitive instruments. They are associated with all types of plate boundaries. Most of the world's earthquakes occur in linear chains along plate boundaries, such as along the west coast of South America. Some earthquakes appear in areas away from plate boundaries, such as in the Midwest of the USA. These earthquakes could still be related to plate movement, as the North American Plate is moving westwards.

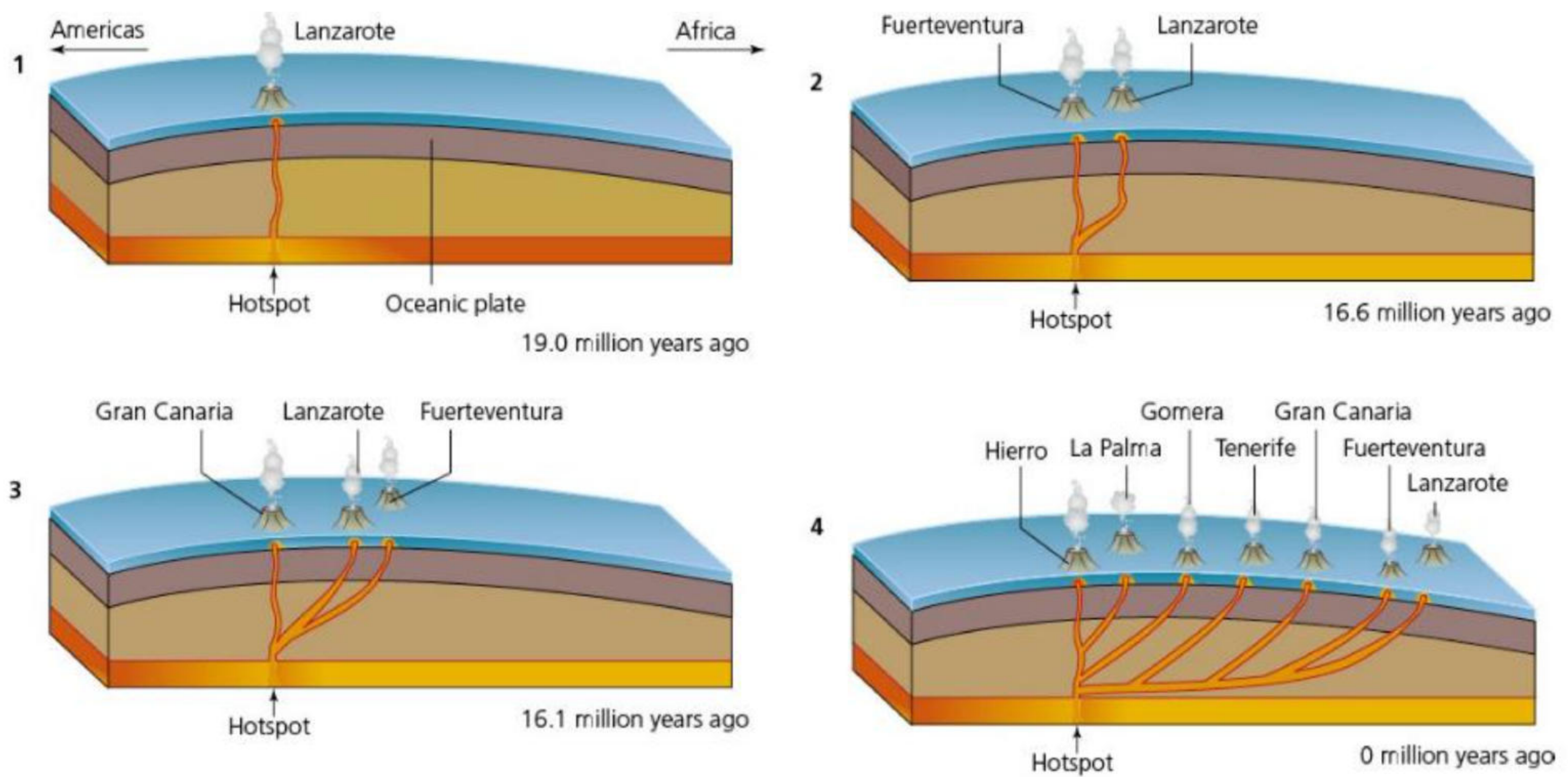
Some earthquakes are the result of human activity. The building of large dams and deep reservoirs increases pressure on the ground. Mining removes underground rocks and minerals, which may cause collapse or subsidence of the overlying materials. Testing of nuclear weapons underground has been known to trigger earthquakes, too.

The location of volcanoes

There are over 1300 active volcanoes in the world, many of them under the ocean (see [Figure 4.6](#)). Three-quarters of the world's active volcanoes are located in the 'Pacific Ring of Fire', the area around the Pacific Ocean. Good examples include Mount Pinatubo (Philippines), Krakatoa (Indonesia) and Popocatépetl (Mexico). These volcanoes are related to plate boundaries, notably convergent/destructive plate boundaries (for example Mount St. Helens in the USA and Soufrière in Montserrat in the Caribbean, see [Figure 4.8](#)) and divergent/constructive boundaries (for example Eldfell volcano on Heimaey, Iceland). The continuing eruption of Soufrière in Montserrat occurs at the boundary of the North American and Caribbean Plates. Some volcanoes, such as Mauna Loa and Kīlauea in Hawaii, and Teide on Tenerife, are located over hotspots. These are isolated plumes of rising magma that have burned through the crust to create active volcanoes ([Figure 4.7](#)).



▲ **Figure 4.6** World distribution of volcanoes and earthquakes



▲ **Figure 4.7** Volcanic activity associated with hotspots



▲ **Figure 4.8** The impacts of the Soufrière Hills volcano, Montserrat

Activities

- 1 Describe the global distribution of earthquakes as shown on [Figure 4.6 \(page 91\)](#).
- 2 Explain why some volcanoes occur in locations away from plate boundaries.